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Received 22 June 2023
Revised 11 September 2023
21 November 2023
7 February 2024
Accepted 16 April 2024

A novel ChatGPT-based multimodel framework for tourism review mining: a case study on China's five sacred mountains

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Abstract

Purpose – This study aims to introduce an innovative framework for mining tourism reviews that not only excels in sentiment analysis accuracy but also prioritizes user-friendly design for enhanced usability.

Design/methodology/approach – Online reviews of China's Five Sacred Mountains were analyzed using an integrated methodology. Sentiment analysis was performed using ChatGPT, bidirectional encoder representations from transformers (BERT) and convolutional neural networks, with ChatGPT demonstrating superior performance. Latent Dirichlet allocation extracted key attributes. Models including importance–performance analysis (IPA), asymmetric impact-performance analysis (AIPA) and importance–performance competitor analysis (IPCA) then synthesized findings.

Findings – The results demonstrate that ChatGPT outperforms both machine learning and lexicon-based models in sentiment recognition, exhibiting performance comparable to that of the BERT model. In the case study, integrating sentiment analysis outcomes with IPA reveals deficiencies in both topics and attributes. Moreover, the synergistic combination of IPA, AIPA and IPCA furnishes actionable recommendations for resource management and enables nuanced monitoring of sustainability attributes.

Practical implications – Leveraging this framework in conjunction with the ChatGPT platform for application development can bring practical convenience to the tourism industry. It supports sentiment

Journal of Hospitality and Tourism Technology
Vol. 15 No. 4, 2024
pp. 592-609
© Emerald Publishing Limited
1757-9880
DOI 10.1108/JHTT-06-2023-0170

analysis, topic categorization and opinion mining. Equipped with monitoring capabilities, it provides valuable insights for sustainable improvement, aiding managers in formulating effective marketing strategies.

Originality/value – This research develops a novel multimodel framework integrating various ML/DL techniques and business models in a synergistic way. It provides an innovative and highly accurate yet simple approach to tourism review mining and enhances accessibility of advanced artificial intelligence for sustainable tourism monitoring, addressing limitations of prior methods.

Keywords Tourism reviews, Sentiment analysis, ChatGPT, Deep learning models, Opinion mining, Sustainable tourism

Paper type Research paper

基于ChatGPT的旅游评论挖掘的新型多模型框架：以中国五岳为例

摘要

研究目的 – 本研究旨在引入一种创新的框架，用于挖掘旅游评论，不仅在情感分析准确性方面表现出色，而且还优先考虑用户友好设计，以提升可用性。

研究方法 – 本研究使用综合方法分析了中国五岳的在线评论，使用ChatGPT进行情感分析。LDA提取了关键属性。然后，包括IPA、AIPA和IPCA在内的模型综合了研究结果。

研究发现 – 结果表明，ChatGPT在情感识别方面优于机器学习和基于词典的模型，表现与BERT模型相当。在案例研究中，将情感分析结果与IPA结合起来揭示了主题和属性的不足。此外，IPA、AIPA和IPCA的协同组合为资源管理提供了可行的建议，并实现了对可持续属性的细致监控

实践意义 – 结合ChatGPT平台在应用开发中利用该框架可以为旅游业带来实际便利。它支持情感分析、主题分类和意见挖掘。配备了监控功能，为可持续改进提供了宝贵的见解，帮助管理者制定有效的营销策略。

研究创新 – 本研究开发了一种新颖的多模型框架，将各种ML/DL技术和商业模型以协同方式整合在一起。它提供了一种创新而高度准确但简单的方法，用于旅游评论挖掘，并提升了高级AI的可访问性，以实现可持续旅游监测。

关键词：旅游评论、情感分析、ChatGPT、深度学习模型、意见挖掘、可持续旅游

文章类型 研究型论文

1. Introduction

Online review mining has become increasingly important in tourism research (Ahani *et al.*, 2019) but faces significant challenges. Traditional machine learning algorithms and lexicon-based methods have limitations, often producing poor sentiment analysis results (Xu and Song, 2023). These limitations make the extracted opinions less useful for practical business analysis.

Advanced deep learning models provide notable improvements in sentiment analysis (Dominic *et al.*, 2023); however, their complexity requires a high level of artificial intelligence (AI) expertise for effective implementation. Meanwhile, There is a noticeable paucity of research focused on leveraging big data and AI for sustainable tourism monitoring and development (Balas and Abson, 2022). This scarcity is primarily attributed to the computational and programming resource constraints inherent to the deployment of AI methodologies. Taking China's developed mountainous tourist areas as a case study provides valuable context for this issue. With mountainous terrain comprising 70% of China's landmass, these regions have historically faced high poverty rates (CCTV News, 2016). The development of mountainous tourist areas is spurred primarily by their abundant natural or cultural heritage resources. Moreover, these areas necessitate local job creation within sensitive ecological environments (UNWTO, 2020), amplifying the demand for sustainable development.

Given these multifaceted challenges, there is an urgent need to develop a highly accurate, yet user-friendly, online review mining methodology that not only excels in sentiment analysis but also circumvents computational and resource limitations. This will not only advance the scope of business analytics in the tourism sector but also contribute to the larger objective of sustainable tourism monitoring.

In response, this study introduces an integrated framework that combines AI techniques with multimodel approaches, as applied to a case study that analyzes review data from mountainous tourist areas. Our approach seeks to enhance service quality in the tourism sector through a more nuanced understanding of tourist preferences, built upon the analytical power of deep learning models like ChatGPT. These models capitalize on Transformer architectures to exploit large volumes of unlabeled data and exhibit self-attention mechanisms for contextual understanding, setting them apart from traditional methods.

Our data set comprises online reviews of China's most prominent mountain ranges, the Five Sacred Mountains (Sacred Mountains of China, 2023), as shown in Figure 1.

Source: Created by author

Figure 1.
Location map of the five sacred mountains in China

The framework is structured into three key components. Initially, it uses a deep learning model, predominantly ChatGPT, to conduct sentiment analysis and identify sentiment polarities within online reviews. Subsequently, the Latent Dirichlet Allocation (LDA) serves as a pivotal mechanism for the extraction and segmentation of topics, alongside the computation of key element weights. Finally, the integration of established business modes – importance–performance analysis (IPA), importance–performance competitor analysis (IPCA) and asymmetric impact–performance analysis (AIPA) – with Python, facilitates a comprehensive extraction and nuanced analysis of tourist feedback. This approach strategically extracts and analyzes entities' strengths and weaknesses, identifies critical feedback and leverages targeted data insights to develop actionable management strategies.

2. Literature review

2.1 Evolution of sentiment analysis models

Recent research has explored various sentiment analysis models across domains. Bidirectional encoder representations from transformers (BERT) consistently achieve high accuracy, outperforming lexicon methods like Valence Aware Dictionary and sEntiment Reasoner (VADER) (Catelli *et al.*, 2022; Ramalingam, 2022) and classifiers like Convolutional Neural Networks (CNN) (Benítez-Andrades *et al.*, 2022). While BERT excels on benchmarks, transformer models like ChatGPT also show promise in initial studies, providing coherent explanations (Zhang *et al.*, 2023) and competitive accuracy on certain tasks (Wang *et al.*, 2023). BERT and ChatGPT each offer unique advantages depending on the application. BERT is often favored for its superior contextual understanding, whereas ChatGPT serves as a versatile, generalist model (Amin *et al.*, 2023). Overall, BERT and new approaches like ChatGPT leads sentiment modeling.

2.2 A Comparative analysis of data mining frameworks in online tourism reviews: methodological advancements and research gaps

Recent advancements in online review mining within tourism and hospitality highlight diverse methodologies and their limitations. Luo *et al.* (2021) used SVM, LDA and IPA for geological park reviews but faced challenges with SVM's limitations in detailed analysis. Albayrak *et al.* (2021) integrated Vader, LDA, IPA and AIPA for theme park reviews, acknowledging the Lexicon-based methods' limitations in assessing unusual word polarity and small data sets. Liu *et al.* (2022) combined dictionary-based methods with LDA for sentiment analysis in Chengdu's food sector, improving lexicon accuracy but struggling with contextual sentiment nuances. Viñán-Ludeña and de Campos (2022) applied BERT for Spanish city reviews, focusing on adjectives but overlooking detailed textual content. Li *et al.* (2022) developed a four-layer neural network for Tokyo restaurant reviews, showing promise but with complexity limiting its broader application. Finally, Li *et al.* (2023) developed a model using CNN to extract semantic meanings from reviews and predict consumer evaluations of restaurants by learning the interactions between consumer preferences and restaurant attributes.

2.3 Applications and research gaps in artificial intelligence and big data for sustainable tourism development

Research on the combined use of AI and big data in sustainable tourism is still in its infancy, though studies have begun using specialized techniques for specific aspects of sustainable tourism. For instance, Giglio *et al.* (2020) used machine learning algorithms for deep data examination, including image recognition and cluster analysis, to explore tourism dynamics

and landmark identification. Periañez-Cristobal *et al.* (2020) used a hybrid machine learning method to examine dynamics like social media reactions. Vázquez Loaiza *et al.* (2019) focused on social network analysis to assess relationships and network structures in tourism.

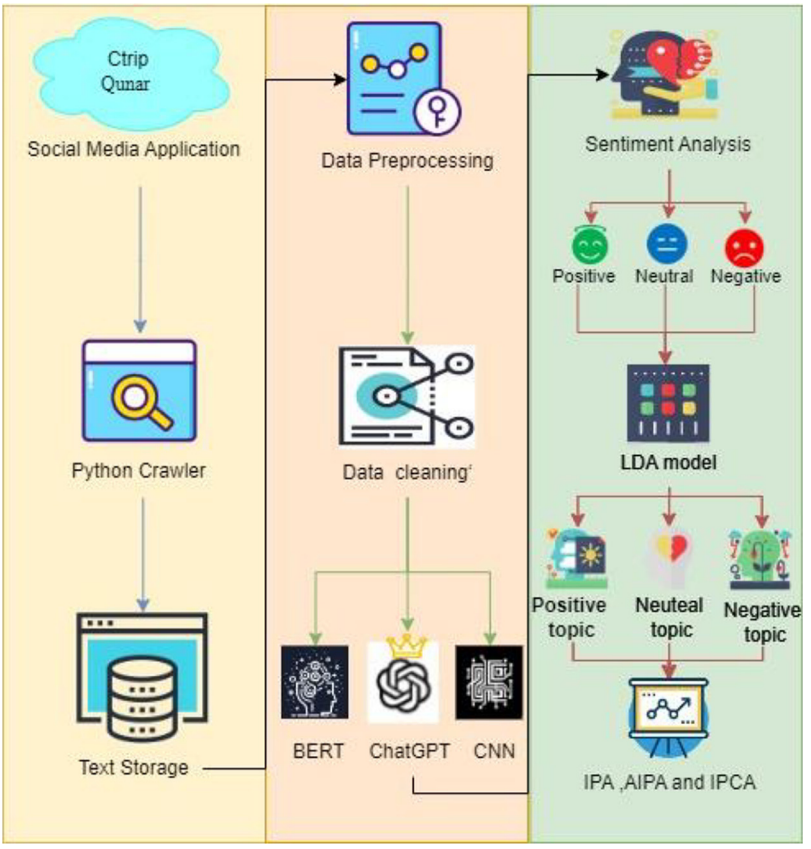
Despite these advancements, as Miller and Torres-Delgado (2023) have already pointed out, there exists a significant gap in the comprehensive integration of these technologies for measuring sustainability. This suggests a pressing need for research that amalgamates AI and big data techniques to measure sustainability indicators in the tourism industry.

3. Research methods

The overall research framework of the paper is shown in Figure 2.

3.1 Data collection and preprocessing

The initial phase involved the collection of data pertaining to the Five Sacred Mountains from Qunar and Ctrip platforms. The data capture was standardized within the time frame



Source: Created by author

Figure 2.
Research framework

of 2015 to 2022, using a Python program for the data collection process. In the following sections, we need to find a powerful sentiment analysis tool.

3.2 Sentiment analysis

3.2.1 Sentiment analysis using ChatGPT API. The purpose of sentiment analysis is to categorize or quantify the emotions expressed in a text. Using the GPT-4-driven ChatGPT API offers significant advantages in sentiment analysis, showcasing capabilities for contextual comprehension and the identification of nuances and sarcasm (Aydın and Karaarslan, 2023). It has been observed to surpass conventional methods in sentiment detection (Holt-Nguyen, 2023) and provides support for large-scale, multilingual projects (Ray, 2023), thereby enhancing decision-making and improving customer satisfaction (Sanghvi, 2023).

The procedure encompasses data preprocessing and API inference. During the preprocessing stage, redundant spaces, punctuation and characters are removed, and the text is converted to lowercase. In addition, word tokenization and normalization are performed. API inference involves meticulous setup adhering to OpenAI's guidelines, preparing prompts for reviews, asynchronous request transmission and parsing responses (OpenAI Platform, 2023); to acquire sentiment scores. The overall algorithm workflow is illustrated in the Algorithm 1.

```
Algorithm 1 Sentiment Analysis of Tourist Reviews using ChatGPT API.
  Step 1: Data cleaning and preparation
  1. Collect reviews.
  2. Clean the text.
  3. Return the prepared text.
  Step 2: Analyzing the sentiment
  4. Set up the ChatGPT API.
  5. Prepare the prompt used for ChatGPT API to process sentiment
     analysis on reviews.
  6. Bundle and process the requests efficiently and call the ChatGPT
     API.
  7. Extract scores from the analysis results of ChatGPT API and store
     the min in a list.
  8. return Sentiment scores.
  Step 3: Evaluation
  9. Obtain accuracy, precision, recall and F1 scores.
```

In the process of constructing the algorithm, we discovered a feature of ChatGPT. We used it to draft the code for sentiment analysis and made some minor modifications to enable it to replace manual coding. The algorithm execution is straightforward, and the coding complexity is low. This sparked our interest in comparing it with other deep learning models. Therefore, in subsection 3.2.2, we only showcase the algorithmic steps of the deep model BERT, exploring whether ChatGPT exhibits user-friendly characteristics through the comparison.

3.2.2 Sentiment analysis using bidirectional encoder representations from transformers model. The BERT model, an accomplished pre-trained language model, offers rich text representations and robust capabilities in sentiment analysis (Devlin *et al.*, 2018) of online reviews of the Five Sacred Mountains .The algorithm workflow is depicted in the Algorithm 2.

```
Algorithm 2 Sentiment Analysis using BERT Model.
  Step 1: Data cleaning and preparation
  1. Collect reviews.
  2. Clean the text.
```

3. Return the prepared text.

Step 2: Model training

4. Set training parameters; load and split dataset into training and validation sets.
5. Encode training and validation sets using Bert Tokenizer, pad texts with lengths less than 128.
6. Define custom dataset class for PyTorch format, return dataset sample.
7. Convert training and validation sets into CuDataset format, and batch data.
8. Build BERT Sequence Classification model, initialize with "bert-base-chinese" transfer to GPU.
9. Define evaluation metrics and model assessment functions for use during training.
10. Train model, input data in batches, record loss values and accuracy.
11. Evaluate model performance on validation set after each epoch, log results.
12. Use AdamW optimizer, CrossEntropyLoss loss function during training.
13. Save model after training.

Step 3: Sentiment classification using BERT model

14. Load data and trained model.
15. Perform sentiment classification, store classifications in specified file.
16. Obtain accuracy, precision, recall and F1 scores.

3.3 Topic feature extraction using latent Dirichlet allocation model

The primary purpose of applying the LDA model in our research is to uncover thematic features and elements from the reviews of the Five Sacred Mountains and allocate the text data into various topics.

LDA can establish a powerful probabilistic mechanism. Its execution steps are illustrated in Algorithm 3.

Algorithm 3 Using LDA for topic feature extraction.

Step 1: Document selection

1. Choose a document based on a predefined probability distribution.

Step 2: Sampling topic distribution

2. Sample the topic distribution for the document from a Dirichlet distribution.

Step 3: Extracting topics for words

3. Use the topic distribution to extract topics for words in the document.
4. Use Gibbs sampling for iterative calculations during this phase.

Step 4: Sampling word distribution

5. Sample the generated word distribution.

Step 5: Word sampling

6. Sample individual words.
7. Calculate the weights of each topic and its associated keywords using the TF-IDF algorithm.

3.4 Business application models: importance–performance analysis, asymmetric impact–performance analysis and importance–performance competitor analysis

After extracting significant insights from deep learning and the LDA model, the study further integrates business models to translate these analyses into actionable strategies and recommendations. It is noteworthy that the accuracy of these business models is paramount to the entire analytical framework.

3.4.1 Importance–performance analysis. The IPA technique aims to assess consumer preferences and satisfaction (Albayrak, 2015), effectively capturing elements of consumer dissatisfaction. Sentiment analysis scores correspond to performance, while LDA weights correspond to importance (Luo *et al.*, 2021).

After using IPA to identify the features of low-performance areas in the Five Sacred Mountains, we will use Python to extract related text content to facilitate the identification of the main causes of low performance.

3.4.2 Asymmetric impact–performance analysis. In traditional IPA, the strength of the impact of topic attributes on performance is not adequately captured. To address this limitation, we use AIPA. In AIPA, the vertical axis represents impact–asymmetry (IA) as proposed by Dueñas *et al.* (2021), while the horizontal axis signifies performance.

Using AIPA allows us to determine which topic has the highest priority for resource allocation. It enables managers to allocate resources based on actual needs, avoiding wastage.

3.4.3 Importance–performance competitor analysis. IPA can offer a broad view of overall performance. When IPA is used in conjunction with IPCA, it allows for a more nuanced analysis of specific features among similar competitors and monitors changes in relevant indicators. The calculation method for IPCA involves two axes: GAP and performance difference (Albayrak, 2015).

GAP reflects the expected value, where a positive GAP indicates that the achieved performance has exceeded expectations, while a negative GAP means the performance has fallen short of expectations.

$$\text{GAP} = \text{performance normalized} - \text{importance normalized}$$

Performance difference, plotted on the horizontal axis, is calculated as the difference between the performance of the focal entity and that of its competitors.

$$\text{Performance difference} = \text{focal performance} - \text{competitor performance}$$

IPCA is divided into four areas as shown in Figure 6, of which the most important area that we monitor is "Urgent Action." The value in this area represents situations where we have not met expected results and have been outperformed by competitors, requiring urgent improvement.

To monitor sustainable indicators, we have designed a method where we use the overall value of the Five Sacred Mountains as a benchmark. We then compare each individual mountain to the overall performance and identify the sustainability indicators that fall within the "Urgent Action" area for monitoring.

4. Findings

4.1 Sentiment analysis model results and discussion

During the execution of sentiment analysis, we conducted a meticulous performance evaluation of various deep learning models, including ChatGPT, BERT and CNN, as well as the control group consisting of SVM, Decision Tree and Vader machine learning models.

The juxtaposition of these performances is documented in Table 1, providing a detailed insight into their comparative strengths and weaknesses.

Particularly noteworthy is the performance of the ChatGPT GPT-4 model, which demonstrated superior efficacy across the majority of the key metrics as delineated in Table 1.

As shown in Table 2, the symbol "★" indicates the complexity of the code, with more "★" suggesting higher complexity. Apart from the computation time, the advantages of using the ChatGPT API are concentrated in several areas. It does not require the purchase of servers or the construction of deep learning frameworks. Considering Code Complexity or use Complexity, it maintains very low requirements. There is no need to intentionally hire high-level AI professionals, which greatly saves on equipment costs and human resources. In addition, GPT-4 uses a powerful pretraining mechanism, allowing it to analyze text from different sources without the need for specific training data. Consequently, the sentiment analysis yielded a distribution comprising 17,512 positive reviews, 3,475 neutral reviews and 924 negative reviews. Besides, the ChatGPT model was used to assign a quantifiable score to each individual review, thereby enhancing the granularity and robustness of the sentiment analysis.

4.2 Latent Dirichlet allocation feature extraction and sentiment scores

After calculating the sentiment scores, the review data was reintegrated with the LDA model for thematic extraction. An intriguing observation was the relative decline in perplexity when the number of topics gravitated around eight. This facilitated the extraction of 8 topic features and 120 topic elements, with 40 elements manually processed and the remaining 80 conserved for ensuing analysis. This is elucidated in Table 3.

The tabular structure portrays the identified features (topics) in the first column, juxtaposed with the top 10 words (elements) predominantly linked to each feature. The

Table 1. Performance comparison of sentiment analysis models

Model type	Model	Accuracy	Precision	Recall	F1 Score
Deep learning	ChatGPT 4.0	0.8413	0.6319	0.6115	0.6215
Deep learning	BERT	0.8197	0.6070	0.6025	0.6047
Deep learning	CNN	0.6269	0.5045	0.4657	0.4843
Machine learning	SVM	0.5675	0.4322	0.4255	0.4286
Machine learning	Decision tree	0.5197	0.4070	0.4025	0.4048
Lexicon method	Vader	0.5337	0.4512	0.4417	0.4463

Source: Created by authors

Table 2. Models comparison by attributes

Model type	Model	Server utilization	Pre-trained model used	Self-attention mechanism	Deep learning framework	Code complexity or use complexity
Deep learning	ChatGPT 4.0	Not used	Used	Used	NONE	★☆☆☆☆
Deep learning	BERT	Used	Used	Used	PYTORCH	★★★★☆
Deep learning	CNN	Used	Not used	Not used	PYTORCH	★★★★☆
Machine learning	SVM	Used	Not used	Not used	NONE	★★★☆☆
Machine learning	Decision tree	Used	Not used	Not used	NONE	★★☆☆☆
Lexicon method	Vader	Not used	Not used	Not used	NONE	★☆☆☆☆

Source: Created by authors

Attributes	Elements of attributes
Cost-effectiveness	Cost-effective (0.008), fun (0.008), value for money (0.009), interesting (0.008), worthwhile (0.009), five great mountains (0.011), recommended (0.008), high value for money (0.008), amusing (0.007), budget (0.008)
Cultural heritage	Shaolin temple (0.011), culture (0.008), three sovereigns and five emperors (0.011), history (0.009), performance (0.008), martial arts (0.017), pagoda forest (0.012), Buddhism (0.009), famous mountains (0.007), temple (0.006)
Special landscapes in mountain tourism	Sunrise (0.019), weather (0.018), summit (0.017), sea of clouds (0.012), rain (0.012), ice and snow (0.011), canyon (0.011), sunlight (0.010), rime (0.009), snow (0.008)
Hiking methods	Cable car (0.025), cableway (0.018), sightseeing bus (0.016), hiking (0.016), physical strength (0.015), time (0.015), on the car (0.013), walking (0.011), on the road (0.010), arrival (0.010)
Ticketing and tourist services	Ticket (0.019), queuing (0.018), ticketing (0.017), parking (0.012), management (0.012), pricing (0.011), transportation (0.010), parking lot (0.010), Ctrip (0.009), online ticketing (0.009)
Scenic spot	Wuyue mountains (0.020), scenic area (0.020), scenic spot (0.017), Jade Emperor Peak (0.012), Waterfall (0.011), Dalong (0.009), summer resort (0.009), night view (0.008), Wulaofeng (0.008), river (0.007)
Tour guide	Tour guide (0.028), service (0.023), scenic area (0.016), explanation (0.016), satisfaction (0.015), happiness (0.014), experience (0.013), scenic spot (0.011), recommendation (0.011), Ctrip (0.009)
Environment and hygiene	Scenery (0.043), garbage (0.017), beautiful (0.014), mountain breeze (0.013), environment (0.012), clean (0.012), graceful (0.010), air (0.010), picturesque (0.010), nature (0.010)

Source: Created by authors

Table 3. Topic features and elements extracted by LDA model

accompanying weight in parentheses symbolizes the significance of each element within its corresponding feature.

Post extraction, the ChatGPT model was engaged to ascertain corresponding sentiment scores for each topic. This duality of analysis yielded average scores for each topic, encapsulating various dimensions such as cost-effectiveness, cultural heritage and special landscapes in mountain tourism. This granular insight plays a crucial role in deciphering the positive and negative sentiments allied with each identified feature.

Following this in-depth extraction and scoring process, the outcomes were harnessed to shape the IPA model, a renowned tool for assessing consumer satisfaction and preferences. The IPA model in our study context synthesized performance aspects from customer reviews scores from ChatGPT recognition, while the importance dimension was signified by weights allocated to topics through LDA calculations. This multifaceted analysis forged a pathway for our rigorous investigation into the facets needing enhancement and systematic understanding.

4.3 Importance–performance analysis

In this section, IPA plays the role of the initial monitoring step to assess the overall deficiencies in the destination. As shown in Figure 3, the focus of the IPA is on the "Concentrate Here" area, which identifies the topics that require urgent improvement. From the graph, it is evident that "Hiking methods" and "Ticketing and tourist services" are the current deficiencies that need urgent improvement. Meanwhile, in the "Low Priority" area,

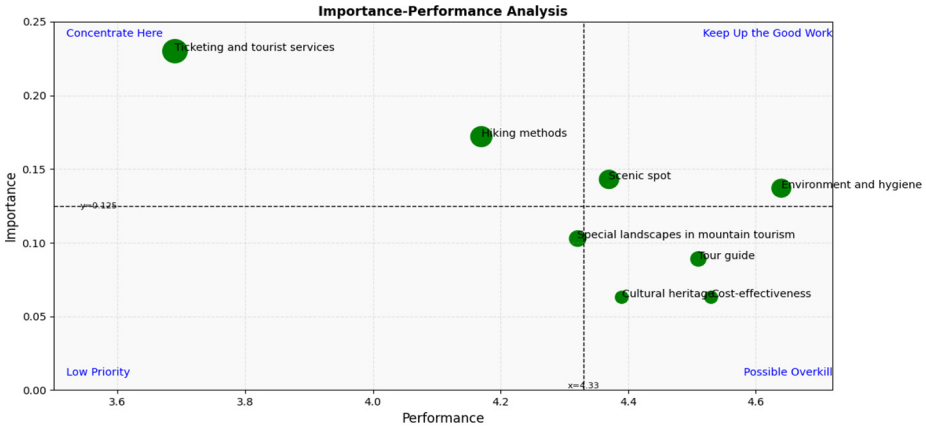


Figure 3.
IPA quadrants for topic attributes

Source: Created by author

"Special Landscapes in Mountain Tourism" also requires improvement but is of lesser importance compared to the aforementioned two topics.

Next, we designed a technique to more finely monitor and display the issues existing in low-performance areas. We expanded the IPA to the "elements" level, as shown in Figure 4. It is clear that Queuing, Ticketing, Buy tickets, Cable car, Sightseeing bus and Management need special attention. Following this, we can directly extract comments related to these elements, analyze them, and identify the shortcomings.

We further crafted a Python script capable of autonomously extracting negative reviews correlated to attributes in the IPA's "Concentrate Here" area. Focusing on "Ticketing," an attribute nestled within this quadrant, we catalog a small subset of corresponding negative remarks in Table 4.

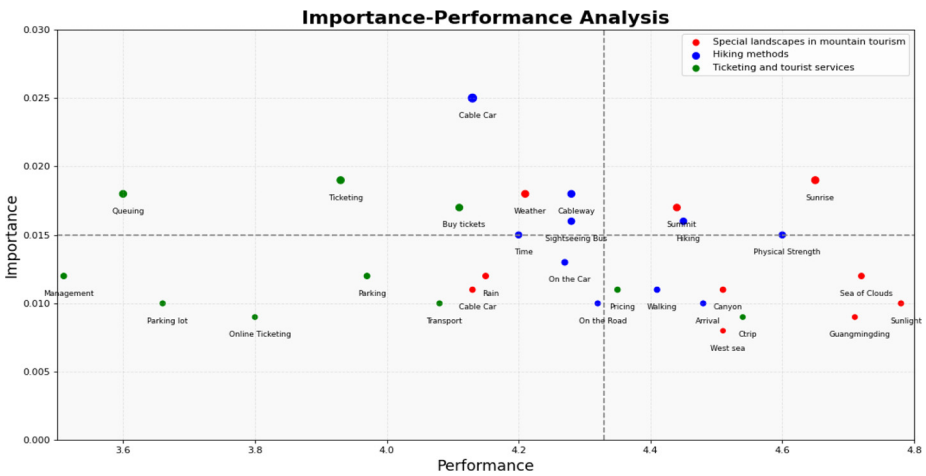


Figure 4.
Distribution of low performance topic-related elements

Source: Created by author

Topic	Content	Name of mountain	Raw scores	ChatGPT score
Ticket	I purchased two tickets online well in advance, but upon reaching the scenic area, I could only collect one ticket. I had to queue and spend an additional 230 yuan to buy another ticket at the entrance, which was a significant waste of time. I am very dissatisfied	Mount Heng (North)	4	1
	The ticket window at the scenic area still doesn't support ticket collection. I had to line up twice to buy tickets at the innermost window, wasting 40 min. Moreover, these tickets are non-refundable. It's advisable to avoid traveling during peak tourist seasons as ticket collection is too challenging	Mount Tai	3	1

Source: Created by authors

Table 4. Negative comments example in the "concentrate here" area of the IPA

From Table 4, salient observations surface. Visitors articulate substantial discontentment with the onerous process of queuing for paper tickets, even after online purchase, an issue demanding immediate management intervention.

More importantly, through manually reviewing the raw text and scores of thousands of online reviews and comparing them with ChatGPT scores, we discovered that ChatGPT is capable of assigning negative scores to certain customer comments that were initially rated as positive or neutral but conveyed negative sentiments in terms of textual meaning. This ability establishes the foundation for the development of an advanced tourism monitoring system.

Next, we will use IPA as the basis to design a resource allocation plan for sustainability and implement monitoring based on sustainability indicators.

4.4 Asymmetric impact–performance analysis

As previously discussed in subsection 3.4.2, we propose to address the limitations of IPA by using AIPA to better understand the impact of various attributes on performance. The key element in AIPA is the acquisition of IA. In the context of big data, the computation of IA necessitates the use of sentiment analysis tools. Several insights were obtained during the calculation process.

The process is as follows:

(1) Segmentation and scoring: Initially, reviews are broken down into individual sentences. Subsequent to this, sentiment scores for each sentence are calculated. Conventionally, the calculation of IA has predominantly relied on lexicon-based methods (Hu *et al.*, 2020). However, as evidenced in subsection 4.1 of our study, the sentiment analysis capabilities afforded by ChatGPT notably surpass those of existing mainstream lexicon-based instruments. Table 5 serves as an illustrative example of the topic-specific scores afforded by ChatGPT.

(2) Coefficient analysis: Here, reward and penalty coefficients are calculated for each attribute. Dummy variables representing the eight attributes are analyzed through regression. The equation for IA is given as:

$$\text{OCS(OverallCustomer Satisfaction)} = \text{C(constant)} + \text{reward index} * \text{reward dummy} + \text{penalty index} * \text{penalty dummy} + \text{err}$$

$$\text{IR (Attribute impact} - \text{range onOCS)} = \text{reward index} + \left|\text{penalty index}\right|$$

$$\text{IA (Impact} - \text{asymmetry)} = \text{reward index /IR} - |\text{penalty index /IR}|$$

(Hu *et al.*, 2020)

When calculating the reward index, all positive emotions are encoded as 1, and all others are encoded as 0. For the penalty index, all negative emotions are encoded as 1, while all others are encoded as 0. Based on this encoding, a virtual regression is performed on the OCS to obtain the aforementioned two indices.

Matrix segmentation: IA values of −0.1 and +0.1 segment the matrix into three areas, following Albayrak *et al.* (2021)'s study.

AIPA categorizes low-performance features into three distinct types:

Basic factors: Features in this area, if low-performing, will reduce satisfaction but are difficult to improve satisfaction if they perform well.

Performance factors: Features in this area will affect changes in satisfaction regardless of high or low performance.

Excitement factors: Features in this area, if high-performing, will lead to increased satisfaction but are unlikely to reduce satisfaction if they perform poorly.

As shown in Figure 5, among the factors exhibiting poor performance, "Ticketing and tourist services" and "Special landscapes in mountain tourism" are categorized as Basic factors. In the case of "Hiking method," it falls under the Excitement factors category. As described in subsection 3.4.2, if managers need to prevent a further decline in satisfaction, they should prioritize investing resources to improve Basic factors. On the contrary, if there is a need to quickly elevate the OCS, resources should be allocated to enhance Excitement factors. Therefore, managers can adjust the OCS based on the actual needs at the time. AIPA can offer valuable recommendations for resource management. In addition, while this only shows an overall analysis for the Five Sacred Mountains, one could separately obtain AIPA values for each individual scenic area and combine them with the IPCA results to form a sustainable resource allocation and monitoring loop.

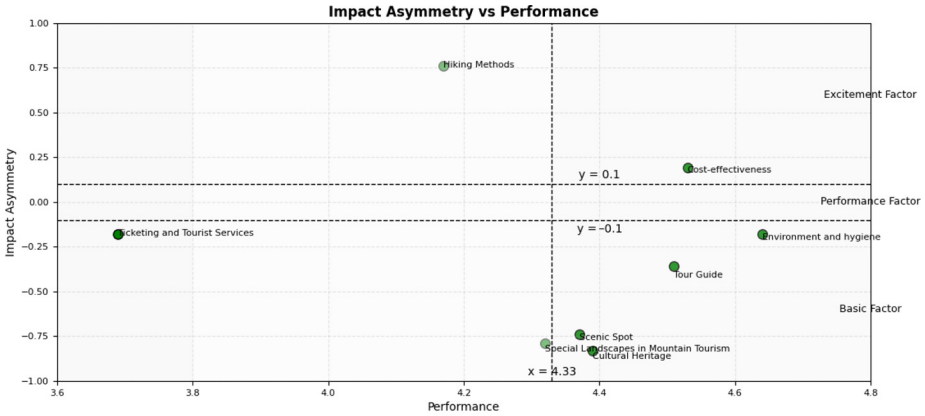
4.5 Importance–performance competitor analysis

IPCA is commonly used in the tourism industry for competitive analysis among similar destinations. We have leveraged it in conjunction with IPA to design a sustainability

Table 5. ChatGPT-assessed sentence scores by topic

Reviews rating	Topics and sentiment scores							
	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8
3				However, the cable car had a long queue, so I ended up walking, which was quite exhausting.2				The mountain is clean, and there is no garbage in sight. 5

Source: Created by authors



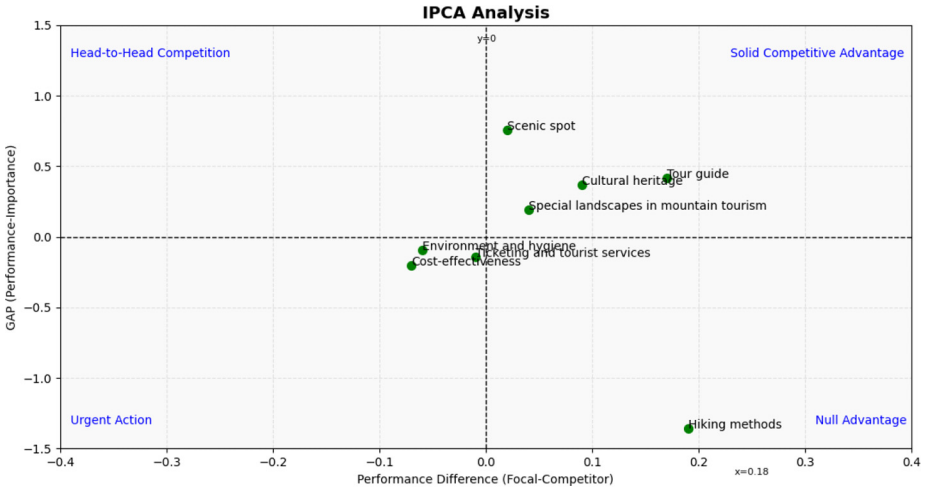
Source: Created by author

Figure 5. Impact asymmetry versus performance

indicator monitoring system. For example, we aimed to understand a typical sustainability feature, specifically focusing on "environmental and hygiene."

We compared the comprehensive values of the Five Sacred Mountains with their respective IPA scores. As depicted in Figure 6, our analysis revealed a surprising discovery: Mount Tai, recognized as a dual UNESCO World Heritage site, a global geopark and a 5A-class scenic area in China, exhibited the poorest performance in "environmental and hygiene," surpassing our previous understanding.

Therefore, we used Python to reverse–extract negative comments related to the "environmental and hygiene" characteristics and elements of Mount Tai. A small subset of these extracted reviews is presented in the Table 6. The extraction results indicate the



Source: Created by author

Figure 6. IPCA analysis of Mount Tai

frequent presence of a large amount of garbage at the summit of Mount Tai. Managers of Mount Tai need to formulate a sustainable management plan to address this issue.

From the comprehensive analysis of monitoring results from IPA, AIPA and IPCA, it is evident that improvements are needed across the Five Sacred Mountains, particularly in areas related to "Hiking methods" and "Ticketing and tourist services." From a monitoring perspective, prioritizing higher resource allocation to "Ticketing and tourist services," as a fundamental factor, is crucial to prevent a rapid decline in overall satisfaction levels. Despite the generally commendable standards in "Environment and Hygiene" across the Five Sacred Mountains, the sustainability metrics of Mount Tai fall below accepted norms, which is deemed unacceptable given its distinguished status.

5. Discussion and conclusion

5.1 Conclusions

This study, by updating the existing framework for tourism reviews, showcases robust sentiment analysis performance. ChatGPT achieves an accuracy of 0.8413 and an F1 score of 0.6215, performing comparably to BERT and surpassing traditional methods. Another advantage of this framework is its user-friendly nature, with low requirements for computational resources and technical expertise. Using this framework, we identified eight topic features and 80 factors. Integrated with a business model, we reverse-extracted insights from reviews, revealing pressing issues related to ticketing and hiking methods for the Five Sacred Mountains. Resource allocation solutions were proposed. In the comparative analysis of the Five Sacred Mountains, sustainability concerns were identified, particularly for Mount Tai, where waste management issues at its summit are significantly underestimated, posing a challenge to its UNESCO World Heritage status. The results indicate that the entire framework, with excellent performance, extracted improvement suggestions for the Five Sacred Mountains from the comments.

5.2 Theoretical implications

Our research has two main theoretical contributions. First, as the field of AI Natural Language Processing has entered the era of Large Language Models (LLM), existing literature in the tourism domain predominantly combines tourism and ChatGPT for qualitative analysis. In contrast, our framework focuses on the application of LLM, updating the methodology for analyzing tourism reviews and filling a theoretical gap in the field. Second, by selecting remote areas as the research focus, this study validates that the use of LLM frameworks in the tourism domain can alleviate resource constraints. Even in the absence of high-level AI professionals and computational power, valuable information can

Table 6. Negative comments examples of the IPCA analysis for Mount Tai

Topic	Content	Raw scores	ChatGPT score
Environment and hygiene	The scenery of Mount Tai is very beautiful, but I was astonished to see a lot of garbage on the summit	1	2
	There is an excessive amount of garbage on the climbing steps. Where are the relevant departments? An explanation for this is warranted	1	1
	The environmental hygiene on the summit of Mount Tai is extremely poor; garbage is everywhere	1	1

Source: Created by authors

be accurately extracted from textual data and tourist satisfaction. Literature on similar sustainable tourism applications is relatively scarce.

5.3 Practical implications

This is a tool for long-term monitoring of tourist satisfaction. OpenAI has introduced a customized version of ChatGPT, enabling anyone to design third-party programs to connect with ChatGPT and build plugins for specific purposes. Developers can package the LDA model, business models and required corresponding Python tools into a third-party program, integrate it with ChatGPT and store it in the cloud. Subsequently, ChatGPT acts directly as the API caller, invoking APIs defined by developers for use by enterprise managers. It is important to emphasize that even without fine-tuning by plugin developers, OpenAI continually updates ChatGPT, enhancing the framework's performance over time. As time progresses, managers can dynamically obtain more accurate data.

Managers can regularly collect vast amounts of unlabeled review data from different platforms, involving different destinations such as hotels, restaurants and attractions. This data can be imported into the plugin for analysis, extracting key themes and factors. The tool not only allows destinations to periodically compare their own satisfaction features but also facilitates aspect-based comparisons with similar destinations, such as comparing identical features among hotels of the same brand. Managers can reverse-extract specific comments based on the weights of different features and the intensity of sentiments, enabling detailed improvements to be made on managerial shortcomings. After improving management strategies, collect relevant comments again after a period and import them into the plugin to assess the effectiveness of the strategies, achieving long-term and fine-grained monitoring.

5.4 Research limitations and future prospects

Several limitations exist in our study. First, our research is constrained by the sample size and future studies should aim to incorporate larger data sets for a more comprehensive analysis. Second, while the framework emphasizes sustainability, it serves only as an initial attempt to merge sustainability metrics with AI. Finally, ChatGPT's limitations in analysis speed – despite recent API speed improvements by OpenAI – along with its restricted availability in some countries, necessitate further enhancements.

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